BMW AND ENFLO TYPE IN METRIC SPACES

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Abstract. We prove that every metric space with Bourgain–Milman–Wolfson type $p \in (1, 2]$ has Enflo type q for every $1 < q < p$. In particular, nontrivial BMW type implies nontrivial Enflo type.

1. Introduction

Let (X, d) be a metric space. For $n \geq 1$, let $Q_n = \{-1, 1\}^n$ with the uniform probability measure, and write $[n] = \{1, \dots, n\}$. For $A \subseteq [n]$, let ε^A be obtained from ε by reversing the coordinates in A. In particular, $\varepsilon^{[n]} = -\varepsilon$, and we write $\varepsilon^{(i)} = \varepsilon^{\{i\}}$. Every expectation on a cube below is normalized.

For $1 \leq p \leq 2$, the space X has *Bourgain–Milman–Wolfson (BMW) type* p [1] with constant $B < \infty$ if

$$\mathbb{E} d(f(\varepsilon), f(-\varepsilon))^2 \leq B^2 n^{2/p-1} \sum_{i=1}^{n} \mathbb{E} d(f(\varepsilon), f(\varepsilon^{(i)}))^2 \tag{1.1}$$

for every integer $n \geq 1$ and every function $f : Q_n \to X$. It has *Enflo type* q, $1 \leq q \leq 2$, with constant $T < \infty$ if

$$\mathbb{E} d(f(\varepsilon), f(-\varepsilon))^q \leq T^q \sum_{i=1}^{n} \mathbb{E} d(f(\varepsilon), f(\varepsilon^{(i)}))^q \tag{1.2}$$

for every integer $n \geq 1$ and every function $f : Q_n \to X$. In both definitions, f is arbitrary; injectivity is not assumed. A space has *nontrivial* BMW type or Enflo type if the corresponding inequality holds with a finite constant for some exponent greater than one. We denote the optimal constants at exponent r by $B_r(X)$ and $E_r(X)$, respectively, allowing the value ∞.

Bourgain, Milman, and Wolfson [1] proved that a metric space has nontrivial BMW type if and only if it does not contain bi-Lipschitz copies of all Hamming cubes with uniformly bounded distortion. Nontrivial Enflo type also excludes such copies. Whether this is the only obstruction to nontrivial Enflo type was recorded as open in [7, Section 2.4] and [2, Introduction, p. 13]. Equivalently, the question is whether nontrivial BMW type forces nontrivial Enflo type.

Our main result gives this implication at every smaller exponent.

Theorem 1.1. *Let $1 < p \leq 2$. Every metric space with BMW type p has Enflo type q for every $1 < q < p$.*

The implication from Enflo type p to BMW type q for every $1 < q < p$ is attributed to Pisier [8] by Prassidis and Weston [9, Section 4]. Proposition 2.3 gives the implication at the same exponent, with $B_p(X) \leq (E_p(X)^2 + 1)^{1/2}$. We have not located an earlier explicit statement of this implication for arbitrary metric spaces when $1 < p < 2$, and make no claim of priority for this observation.

Together with Theorem 1.1, this shows that the supremal BMW and Enflo type exponents of every metric space coincide. In particular, nontrivial BMW type and nontrivial Enflo type are equivalent, which answers the qualitative question above.

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This equivalence also gives a finite-union result. Karamanlis and Kravaris [5, Proposition 1.5] proved that nontrivial BMW type is preserved under finite unions of subspaces, and left the corresponding Enflo statement unresolved [5, footnote 2]. Applying the comparison in both directions gives the same finite-union property for nontrivial Enflo type.

For $1 < p < 2$, the range $q < p$ is sharp even for Banach spaces. Ivanisvili, van Handel, and Volberg [4] proved that Rademacher type p and Enflo type p coincide in that setting. Their inequality also gives the comparison between BMW type and equal-norm type proved in Section 5. Combining this comparison with Tzafriri's examples [10] gives the following endpoint failure.

Corollary 1.2. *For every $1 < p < 2$, there exists a Banach space Y_p such that*

$$B_p(Y_p) < \infty, \qquad E_p(Y_p) = \infty,$$

while $E_q(Y_p) < \infty$ for every $1 \le q < p$.

At $p = 2$, the BMW and Enflo inequalities are identical, so the endpoint holds for every metric space.

The proof begins with a scalar Poincaré estimate for partial-flip distances. We apply the BMW inequality to functions obtained by flipping blocks of coordinates and average over the base vertex. A suitable partition yields $T_n \le \theta T_n + C$, where T_n is the finite Enflo constant in dimension n and $\theta < 1$. Absorbing the term θT_n gives the required uniform bound.

Section 2 supplies the lemmas used in the proof in Section 3. Section 4 gives the consequences, and Section 5 treats sharpness.

2. Preliminaries

We first control the fluctuations of distances associated with coordinate flips. The starting point is a scalar Poincaré inequality. For a real-valued function h on Q_n, write

$$\partial_i h(\varepsilon) = \frac{h(\varepsilon) - h(\varepsilon^{(i)})}{2}, \qquad \|h\|_q = (\mathbb{E}|h|^q)^{1/q}.$$

Lemma 2.1. *For $1 \le q \le 2$ and every $h : Q_n \to \mathbb{R}$,*

$$\mathbb{E}|h - \mathbb{E}h|^q \le \sum_{i=1}^n \mathbb{E}|\partial_i h|^q = 2^{-q} \sum_{i=1}^n \mathbb{E}|h(\varepsilon) - h(\varepsilon^{(i)})|^q. \tag{2.1}$$

Proof. Let $\mathcal{F}_i$ be generated by $\varepsilon_1, \dots, \varepsilon_i$, with $\mathcal{F}_0$ trivial, and put

$$M_i = \mathbb{E}[h \mid \mathcal{F}_i], \qquad \Delta_i = M_i - M_{i-1}.$$

Conditionally on $\mathcal{F}_{i-1}$, the variable Δ_i takes two opposite values with equal probabilities. For real u, v,

$$\frac{|u+v|^q + |u-v|^q}{2} \le (u^2 + v^2)^{q/2} \le |u|^q + |v|^q. \tag{2.2}$$

The first inequality is the comparison of the L_q and L_2 norms on a two-point probability space, and the second follows from $q/2 \le 1$. Applying (2.2) conditionally to $M_i - M_0 = (M_{i-1} - M_0) + \Delta_i$ and iterating gives

$$\mathbb{E}|h - \mathbb{E}h|^q \le \sum_{i=1}^n \mathbb{E}|\Delta_i|^q. \tag{2.3}$$

Moreover, conditional averaging over the unrevealed coordinates gives

$$\mathbb{E}[h(\varepsilon^{(i)}) \mid \mathcal{F}_i] = M_i(\varepsilon^{(i)}) = 2M_{i-1}(\varepsilon) - M_i(\varepsilon).$$

Thus

$$\Delta_i = \mathbb{E}[\partial_i h \mid \mathcal{F}_i].$$

Conditional Jensen yields $\mathbb{E}|\Delta_i|^q \le \mathbb{E}|\partial_i h|^q$. Substitution in (2.3) proves (2.1). □

Given $f : Q_n \to X$ and $1 \le q \le 2$, set

$$\begin{aligned} e_i(\varepsilon) &= d(f(\varepsilon), f(\varepsilon^{(i)})), \qquad S = \sum_{i=1}^{n} \mathbb{E} e_i^q, \\ b_A(\varepsilon) &= d(f(\varepsilon), f(\varepsilon^A)), \quad \mu_A = \mathbb{E} b_A. \end{aligned} \tag{2.4}$$

In particular, $b_\varnothing = 0$. All these quantities are finite because the domain of f is finite.

Lemma 2.2. *For every fixed $A \subseteq [n]$,*

$$\|b_A - \mu_A\|_q^q \le S. \tag{2.5}$$

Proof. The reverse triangle inequality and commutativity of coordinate flips give

$$|b_A(\varepsilon) - b_A(\varepsilon^{(i)})| \le e_i(\varepsilon) + e_i(\varepsilon^A).$$

By convexity and invariance of uniform measure under the flip of A,

$$\begin{aligned} \mathbb{E}|\partial_i b_A|^q &\le \mathbb{E}\left(\frac{e_i(\varepsilon) + e_i(\varepsilon^A)}{2}\right)^q \\ &\le \frac{1}{2}\mathbb{E}\big(e_i(\varepsilon)^q + e_i(\varepsilon^A)^q\big) = \mathbb{E} e_i^q. \end{aligned}$$

Apply Lemma 2.1 to b_A and sum over i. □

Applying the preceding concentration estimate with exponent 2 also gives the reverse implication announced in the introduction.

Proposition 2.3. *For every metric space X and every $1 \le p \le 2$,*

$$B_p(X) \le \big(E_p(X)^2 + 1\big)^{1/2}. \tag{2.6}$$

Proof. It suffices to consider an admissible finite Enflo type p constant T. For $f : Q_n \to X$, put $b = b_{[n]}$ and $S_2 = \sum_i \mathbb{E} e_i^2$. The Enflo inequality, monotonicity of moments on a probability space, and Hölder's inequality give

$$\begin{aligned} (\mathbb{E} b)^2 \le (\mathbb{E} b^p)^{2/p} &\le T^2 \left(\sum_{i=1}^{n} \mathbb{E} e_i^p\right)^{2/p} \\ &\le T^2 \left(\sum_{i=1}^{n} (\mathbb{E} e_i^2)^{p/2}\right)^{2/p} \le T^2 n^{2/p-1} S_2. \end{aligned}$$

Lemma 2.2, applied with exponent 2, yields $\mathrm{Var}(b) \le S_2$. Since $n^{2/p-1} \ge 1$,

$$\mathbb{E} b^2 = (\mathbb{E} b)^2 + \mathrm{Var}(b) \le (T^2 + 1) n^{2/p-1} S_2.$$

Thus $B_p(X) \le (T^2 + 1)^{1/2}$. Taking the infimum over admissible T proves (2.6). □

3. Proof of the main theorem

Proof of Theorem 1.1. Let $B > 0$ be a BMW type p constant for X. Fix $1 < q < p$ and choose an integer $m \geq 2$ such that

$$\theta = Bm^{1/p-1/q} < 1. \tag{3.1}$$

This is possible because $q < p$. Put

$$\kappa = Bm^{1/p-1/2}, \qquad K = m^{1/q}2^{m(1/q-1/2)}.$$

For $n \geq 1$, define

$$T_n = \sup_{\substack{f:Q_n\to X \\ S(f)>0}} \frac{\|b_{[n]}\|_q}{S(f)^{1/q}}, \tag{3.2}$$

with value zero if there are no maps of positive edge energy. The triangle inequality along a coordinate path, followed by Minkowski and Hölder, gives

$$\|b_{[n]}\|_q \leq \sum_{i=1}^{n} \|e_i\|_q \leq n^{1-1/q} S(f)^{1/q}.$$

Thus $T_n < \infty$. Fix $f : Q_n \to X$ with $S > 0$, and put

$$s_i = \frac{\mathbb{E}e_i^q}{S}, \qquad \sum_{i=1}^{n} s_i = 1.$$

For any $A \subseteq [n]$, apply the inequality defining T_n to each face in the coordinates of A, adding unused coordinates to its domain, and average over the coordinates outside A. This gives

$$\mu_A \leq \|b_A\|_q \leq T_n S^{1/q} \left(\sum_{i\in A} s_i\right)^{1/q}. \tag{3.3}$$

For $A = \varnothing$, both sides vanish.

Partition of the coordinates. Assign the coordinates one at a time to a block of smallest current weight, producing a partition $A_1, \ldots, A_m$ of $[n]$. Empty blocks are allowed. For each nonempty A_j, let i_j be the last coordinate assigned to it, and put $C_j = A_j \setminus \{i_j\}$. Just before that assignment, the block had minimum weight among m blocks whose total weight was at most one. Hence

$$\sum_{i\in C_j} s_i \leq \frac{1}{m}.$$

The triangle inequality and invariance under flips, followed by (3.3), imply

$$\mu_{A_j} \leq \mu_{C_j} + \mathbb{E}e_{i_j} \leq S^{1/q}\big(T_n m^{-1/q} + s_{i_j}^{1/q}\big).$$

The distinguished coordinates i_j are distinct, and $2/q \geq 1$. Minkowski in ℓ_2^m therefore yields

$$\begin{aligned}\left(\sum_{j=1}^{m} \mu_{A_j}^2\right)^{1/2} &\leq S^{1/q}\left[T_n m^{1/2-1/q} + \left(\sum_{j:A_j\neq\varnothing} s_{i_j}^{2/q}\right)^{1/2}\right] \\ &\leq S^{1/q}\big(T_n m^{1/2-1/q} + 1\big).\end{aligned} \tag{3.4}$$

Empty blocks contribute zero to these estimates.

Averaging the BMW inequality. For every $\varepsilon \in Q_n$, define $F_\varepsilon : Q_m \to X$ by

$$F_\varepsilon(\sigma) = f(\varepsilon^{U_\sigma}), \qquad U_\sigma = \bigcup_{\{j:\sigma_j=-1\}} A_j. \tag{3.5}$$

Since the blocks partition $[n]$,

$$d(F_\varepsilon(\sigma), F_\varepsilon(\sigma^{(j)})) = b_{A_j}(\varepsilon^{U_\sigma}),$$
$$d(F_\varepsilon(\sigma), F_\varepsilon(-\sigma)) = b_{[n]}(\varepsilon^{U_\sigma}).$$

Apply BMW to F_ε, use $L_1(Q_m) \le L_2(Q_m)$ on its diagonals, and then average in ε. Each fixed translate ε^{U_σ} is uniform, so

$$\mu_{[n]} \le \kappa\, \mathbb{E}_\varepsilon \left(\sum_{j=1}^m \mathbb{E}_\sigma b_{A_j}(\varepsilon^{U_\sigma})^2 \right)^{1/2}. \tag{3.6}$$

To separate the means from the fluctuations, set

$$Z(\varepsilon) = \left(\sum_{j=1}^m \mathbb{E}_\sigma |b_{A_j}(\varepsilon^{U_\sigma}) - \mu_{A_j}|^2 \right)^{1/2}.$$

Minkowski gives

$$\mu_{[n]} \le \kappa \left[\left(\sum_{j=1}^m \mu_{A_j}^2 \right)^{1/2} + \mathbb{E}_\varepsilon Z(\varepsilon) \right]. \tag{3.7}$$

For the $m2^m$ entries indexed by (j,σ), the unnormalized sequence norms satisfy $\|\cdot\|_{\ell_2} \le \|\cdot\|_{\ell_q}$. Keeping the normalization 2^{-m} in $\mathbb{E}_\sigma$, then using Jensen, translation invariance, and Lemma 2.2, we obtain

$$\begin{aligned} \mathbb{E}_\varepsilon Z(\varepsilon) &\le 2^{m(1/q-1/2)} \left(\sum_{j=1}^m \mathbb{E}_{\varepsilon,\sigma} |b_{A_j}(\varepsilon^{U_\sigma}) - \mu_{A_j}|^q \right)^{1/q} \\ &\le 2^{m(1/q-1/2)} (mS)^{1/q} = KS^{1/q}. \end{aligned} \tag{3.8}$$

This finite-dimensional comparison involves m, independently of n.

Absorption. Lemma 2.2 also gives $\|b_{[n]}\|_q \le \mu_{[n]} + S^{1/q}$. Combining (3.4), (3.7), and (3.8), we conclude that

$$\frac{\|b_{[n]}\|_q}{S^{1/q}} \le \kappa m^{1/2-1/q} T_n + 1 + \kappa(1+K) = \theta T_n + 1 + \kappa(1+K).$$

Taking the supremum over f gives

$$T_n \le \theta T_n + 1 + \kappa(1+K).$$

Since $T_n < \infty$ and $\theta < 1$, it follows that

$$T_n \le C_q := \frac{1+\kappa(1+K)}{1-\theta} = \frac{1 + Bm^{1/p-1/2}\big(1 + m^{1/q} 2^{m(1/q-1/2)}\big)}{1 - Bm^{1/p-1/q}}. \tag{3.9}$$

The choice of m and the resulting bound depend only on B, p, q. Thus X has Enflo type q. Maps with $S = 0$ are constant and satisfy the conclusion trivially. □

Remark 3.1. All martingales in the proof are real-valued, and every auxiliary function $F : Q_m \to X$ takes values in X itself. No completeness, geodesic, barycenter, or linear assumption on the target is used. Singleton spaces and maps of zero edge energy satisfy all the conclusions trivially.

4. Consequences

We first express Theorem 1.1 and Proposition 2.3 in terms of supremal exponents. For a metric space X, define

$$p_{\mathrm{BMW}}(X) = \sup\{r \in [1,2] : B_r(X) < \infty\}, \qquad p_{\mathrm{Enflo}}(X) = \sup\{r \in [1,2] : E_r(X) < \infty\}.$$

Both sets contain 1: a coordinate path and the triangle inequality give Enflo type 1, and Minkowski followed by Cauchy–Schwarz gives BMW type 1.

Corollary 4.1. *For every metric space X,*

$$p_{\mathrm{BMW}}(X) = p_{\mathrm{Enflo}}(X).$$

Proof. Proposition 2.3 gives $p_{\mathrm{Enflo}}(X) \le p_{\mathrm{BMW}}(X)$. For every $p > 1$ with $B_p(X) < \infty$, Theorem 1.1 gives $E_q(X) < \infty$ for all $1 < q < p$, so $p_{\mathrm{Enflo}}(X) \ge p$. Taking the supremum over such p gives the reverse inequality. If $p_{\mathrm{BMW}}(X) = 1$, both suprema are already equal to 1. No attainment of either supremum is assumed. □

The equivalence of nontrivial BMW and Enflo type also has a geometric formulation in terms of Hamming cubes. Equip Q_n with the Hamming metric

$$d_H(\varepsilon, \eta) = |\{i \in [n] : \varepsilon_i \neq \eta_i\}|.$$

We say that X *contains the Hamming cubes uniformly* if there is $D < \infty$ such that, for every n, there exist $g_n : Q_n \to X$ and $s_n > 0$ satisfying

$$s_n d_H(\varepsilon, \eta) \le d(g_n(\varepsilon), g_n(\eta)) \le D s_n d_H(\varepsilon, \eta) \qquad (\varepsilon, \eta \in Q_n). \tag{4.1}$$

Corollary 4.2. *For an arbitrary metric space X, the following conditions are equivalent:*

(i) *X has nontrivial BMW type.*
(ii) *X has nontrivial Enflo type.*
(iii) *X does not contain the Hamming cubes uniformly.*

Proof. For (i)⇒(ii), suppose that X has BMW type $p > 1$. Apply Theorem 1.1 with $q = (1+p)/2$.

For (ii)⇒(iii), suppose that X has Enflo type $q > 1$ with a constant $T > 0$. If (4.1) holds in dimension n, each antipodal distance is at least $s_n n$ and each edge length is at most $D s_n$. Applying (1.2) gives

$$(s_n n)^q \le T^q n (D s_n)^q, \qquad \text{hence} \qquad D \ge \frac{n^{1-1/q}}{T}.$$

Thus D cannot be bounded independently of n.

The implication (iii)⇒(i) is the theorem of Bourgain, Milman, and Wolfson [1]; see also [7, Section 2.4]. □

This characterization transfers the finite-union theorem for BMW type to Enflo type.

Corollary 4.3. *Let (X, d) be a metric space and suppose that $X = X_1 \cup \cdots \cup X_k$, where every X_j, equipped with the induced metric, has nontrivial Enflo type. Then X has nontrivial Enflo type.*

Proof. By Corollary 4.2, every X_j has nontrivial BMW type. The finite-union theorem of Karamanlis and Kravaris [5, Proposition 1.5], applied repeatedly, implies that X has nontrivial BMW type. Another application of Corollary 4.2 completes the proof. □

5. Sharpness of the exponent

It remains to examine the endpoint $q = p$. At $p = 2$, the defining inequalities coincide. For $1 < p < 2$, we prove Corollary 1.2 by identifying BMW type p with equal-norm Rademacher type p in Banach spaces. This comparison follows from [4, Theorem 1.2], and Tzafriri's examples [10] then give the required failure of Enflo type p.

For vectors $x_1, \dots, x_n$ in a Banach space X, write

$$R_X(x_1, \dots, x_n) = \left(\mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i x_i \right\|^2 \right)^{1/2}.$$

The space has Rademacher type p if there is a finite constant T such that

$$R_X(x_1, \dots, x_n) \le T \left(\sum_{i=1}^n \|x_i\|^p \right)^{1/p} \tag{5.1}$$

for all finite sequences. By the Kahane–Khintchine inequalities, replacing the second moment by the pth moment gives the same property.

Let $K_p(X)$ be the least constant in

$$R_X(x_1, \dots, x_n) \le K_p(X) n^{1/p} \max_{1 \le i \le n} \|x_i\|. \tag{5.2}$$

Finiteness of $K_p(X)$ is called *equal-norm type p*. By the contraction principle, this is equivalent to requiring (5.1) only for vectors of equal norm. Together with moment equivalence, this identifies the definition with the one in [3, pp. 206–207]. To connect this condition with the BMW inequality, let $A_p(X)$ be the least constant in

$$R_X(x_1, \dots, x_n) \le A_p(X) n^{1/p - 1/2} \left(\sum_{i=1}^n \|x_i\|^2 \right)^{1/2}. \tag{5.3}$$

This formulation is also called equal-norm type in some accounts; see [6, Section 4.3, equation (10)]. We first compare $A_p(X)$ with $B_p(X)$, and then relate $A_p(X)$ to $K_p(X)$.

Proposition 5.1. *For every Banach space X and $1 \le p \le 2$,*

$$A_p(X) \le B_p(X) \le \frac{\pi}{\sqrt{2}} A_p(X). \tag{5.4}$$

Proof. Apply (1.1) to the map $f(\varepsilon) = \sum_i \varepsilon_i x_i$. Its diagonal has length $2\| \sum_i \varepsilon_i x_i \|$, and its ith edge has length $2\|x_i\|$. Cancelling the common factor 4 proves $A_p(X) \le B_p(X)$.

For the converse, let $A < \infty$ be an admissible constant in (5.3). For a Banach-valued map $f : Q_n \to X$, set

$$D_i f(\varepsilon) = \frac{f(\varepsilon) - f(\varepsilon^{(i)})}{2}.$$

Theorem 1.2 of [4], applied to the convex function $x \mapsto \|x\|^2$, gives

$$\mathbb{E}\|f - \mathbb{E}f\|^2 \le \frac{\pi^2}{4} \int_0^\infty \mathbb{E}_{\varepsilon,\delta} \left\| \sum_{i=1}^n \delta_i(t) D_i f(\varepsilon) \right\|^2 \mu(dt). \tag{5.5}$$

Here

$$\mu(dt) = \frac{2\,dt}{\pi\sqrt{e^{2t}-1}}, \qquad \delta_i(t) = \frac{\xi_i(t) - e^{-t}}{\sqrt{1-e^{-2t}}},$$

where the $\xi_i(t)$ are independent signs, independent of ε, with $\mathbb{P}(\xi_i(t)=1) = (1+e^{-t})/2$. Thus $\mathbb{E}\delta_i(t) = 0$ and $\mathbb{E}\delta_i(t)^2 = 1$. The measure μ is a probability measure, since the substitution $u = e^{-t}$ gives

$$\int_0^\infty \mu(dt) = \frac{2}{\pi}\int_0^1 \frac{du}{\sqrt{1-u^2}} = 1.$$

Fix $t > 0$ and deterministic vectors $v_1, \dots, v_n$. Let δ' be an independent copy of $\delta(t)$, and let $\eta_1, \dots, \eta_n$ be independent uniform signs, independent of both copies. Jensen's inequality, symmetry of the independent differences $\delta_i - \delta_i'$, and (5.3) give

$$\begin{aligned}
\mathbb{E}_\delta \left\| \sum_i \delta_i v_i \right\|^2 &\le \mathbb{E}_{\delta,\delta'} \left\| \sum_i (\delta_i - \delta_i') v_i \right\|^2 \\
&= \mathbb{E}_{\delta,\delta',\eta} \left\| \sum_i \eta_i (\delta_i - \delta_i') v_i \right\|^2 \\
&\le A^2 n^{2/p-1} \sum_i \mathbb{E}|\delta_i - \delta_i'|^2 \|v_i\|^2 \\
&= 2A^2 n^{2/p-1} \sum_i \|v_i\|^2.
\end{aligned}$$

Apply this with $v_i = D_i f(\varepsilon)$ for each fixed ε, and then average in (5.5). We obtain

$$\mathbb{E}\|f - \mathbb{E}f\|^2 \le \frac{\pi^2}{2} A^2 n^{2/p-1} \sum_i \mathbb{E}\|D_i f\|^2.$$

Finally, the triangle inequality and invariance under $\varepsilon \mapsto -\varepsilon$ yield

$$\mathbb{E}\|f(\varepsilon) - f(-\varepsilon)\|^2 \le 4\mathbb{E}\|f - \mathbb{E}f\|^2, \qquad 4\|D_i f(\varepsilon)\|^2 = \|f(\varepsilon) - f(\varepsilon^{(i)})\|^2.$$

These estimates give BMW type p with constant $(\pi/\sqrt{2})A$. Taking the infimum over admissible A proves (5.4). If no finite A exists, the upper bound is immediate. □

For $1 < p < 2$, a dyadic decomposition gives the remaining comparison.

Lemma 5.2. *Let $1 < p < 2$ and $c_p = (1 - 2^{-(1/p-1/2)})^{-1}$. Then*

$$K_p(X) \le A_p(X) \le c_p K_p(X). \tag{5.6}$$

Proof. The first inequality follows from $(\sum_i \|x_i\|^2)^{1/2} \le \sqrt{n}\max_i \|x_i\|$. For the second, assume $K_p(X) < \infty$ and reorder the vectors so that $a_i = \|x_i\|$ is nonincreasing. Set $\ell = \lfloor \log_2 n \rfloor$ and

$$I_k = \{2^k, \dots, \min(2^{k+1}-1, n)\}, \qquad 0 \le k \le \ell.$$

Minkowski's inequality and (5.2) give

$$R_X(x_1, \dots, x_n) \le \sum_{k=0}^{\ell} R_X((x_i)_{i\in I_k}) \le K_p(X) \sum_{k=0}^{\ell} 2^{k/p} a_{2^k}.$$

Since $2^k a_{2^k}^2 \le \sum_i a_i^2$, putting $\alpha = 1/p - 1/2 > 0$ yields

$$R_X(x_1, \dots, x_n) \le K_p(X) \left(\sum_i a_i^2 \right)^{1/2} \sum_{k=0}^{\ell} 2^{k\alpha}.$$

The geometric sum satisfies

$$\sum_{k=0}^{\ell} 2^{k\alpha} \le \frac{2^{\ell\alpha}}{1-2^{-\alpha}} \le c_p n^{\alpha}. \qquad \square$$

Combining Proposition 5.1 and Lemma 5.2, we obtain

$$K_p(X) \le B_p(X) \le \frac{\pi c_p}{\sqrt{2}} K_p(X), \qquad 1 < p < 2. \tag{5.7}$$

Thus BMW type p and equal-norm type p are equivalent for Banach spaces in this range, at the same exponent.

Proof of Corollary 1.2. For every $1 < p < 2$, Tzafriri [10] constructed a Banach space Y_p with equal-norm type p but without Rademacher type p; see also [3, pp. 206–207], where this existence statement and the definition of equal-norm type are explicitly recorded. In the notation above, $K_p(Y_p) < \infty$, so (5.7) gives $B_p(Y_p) < \infty$.

If Y_p had Enflo type p with constant T, applying (1.2) to $f(\varepsilon) = \sum_i \varepsilon_i y_i$ and cancelling 2^p would give

$$\mathbb{E}\left\|\sum_i \varepsilon_i y_i\right\|^p \le T^p \sum_i \|y_i\|^p.$$

This is Rademacher type p, a contradiction. Finally, Theorem 1.1 gives Enflo type q for every $1 < q < p$, and Enflo type 1 holds in every metric space. $\square$

For these examples, both supremal exponents in Corollary 4.1 are exactly p. Indeed, BMW type r for some $r > p$ would imply Enflo type p by Theorem 1.1. Hence the BMW exponent is attained at p, whereas the Enflo exponent is not.

Remark 5.3. The absorption in the main proof requires $q < p$. At $q = p$ its coefficient becomes B. If X contains at least two points, the one-dimensional BMW inequality forces $B \ge 1$, so the absorption is no longer available. Corollary 1.2 supplies a separate reason that the endpoint conclusion cannot hold in general for $1 < p < 2$. The converse implication, from Enflo type p to BMW type p, holds by Proposition 2.3. At $p = 2$, the defining BMW and Enflo inequalities are identical: $B_2(X) = E_2(X)$ for every metric space. For Banach spaces, this property is also equivalent to Rademacher type 2 by [4, Theorem 1.1].

References

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